PROJECT REPORT

ON

CMOS INVERTER DESIGN AND SIMULATION USING LTspice

Submitted in partial fulfilment of the requirements of the VLSI course

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1. ABSTRACT

A Complementary Metal-Oxide-Semiconductor (CMOS) inverter is a fundamental building block of digital integrated circuits. It performs the NOT logic operation, producing an output that is the complement of its input. When the input is LOW, the output is HIGH; when the input is HIGH, the output is LOW.

This project presents the design and transient simulation of a CMOS inverter using LTspice. The circuit consists of one PMOS transistor connected to the positive supply and one NMOS transistor connected to ground. Their gates are connected together to form the input, while their drains share the output node. A 5 V supply and a pulsed input switching between 0 V and 5 V are used to demonstrate inverter operation. Transient analysis is used to observe the input and output waveforms. The simplified Level-1 transistor models are for educational demonstration and are not calibrated to a specific semiconductor fabrication process.

2. INTRODUCTION

Very Large Scale Integration (VLSI) technology enables large numbers of transistors to be integrated on a single semiconductor chip. Microprocessors, memories, microcontrollers, communication circuits, and many digital systems are built using VLSI techniques. CMOS technology is widely used because complementary PMOS and NMOS devices implement logic with low static power consumption in ideal stable states.

Digital circuits represent information using logic 0 (LOW) and logic 1 (HIGH). Logic gates perform Boolean operations on these states. An inverter, also called a NOT gate, reverses the logic state of its input. Although simple, the CMOS inverter demonstrates pull-up and pull-down networks, switching behaviour, voltage levels, and power consumption.

This project designs a CMOS inverter at transistor level and verifies its behaviour using LTspice. A time-varying input is applied, and the output waveform is examined to verify inversion.

3. PROBLEM STATEMENT

A digital system requires a circuit that produces the complement of an input logic signal. The circuit must produce a HIGH output when its input is LOW and a LOW output when its input is HIGH. The project constructs a transistor-level CMOS inverter, configures a supply and input source, runs transient simulation, and verifies the relationship between input and output. Correct transistor connections, grounding, model selection, and simulation settings are necessary for meaningful results.

4. OBJECTIVES

- Understand the operating principles of NMOS and PMOS transistors.
- Study complementary switching in CMOS technology.
- Design a CMOS inverter using one PMOS and one NMOS transistor.
- Construct and simulate the circuit in LTspice.
- Configure a 5 V supply and pulsed input source.
- Perform transient analysis and observe input/output waveforms.
- Verify the NOT-gate truth table.
- Understand the role of CMOS in digital integrated circuits.

• Establish a foundation for more complex CMOS logic-gate design.

5. SCOPE OF THE PROJECT

The project covers transistor-level circuit construction, source configuration, transient simulation, waveform observation, and functional verification. It is a simulation-based educational exercise and does not include physical chip fabrication, semiconductor layout, design-rule checking, or laboratory measurement. Further work could include DC transfer-characteristic analysis, propagation-delay measurement, power analysis, transistor sizing, and layout design.

6. THEORETICAL BACKGROUND

A MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) is a voltage-controlled semiconductor device used as a switch or amplifier. Its terminals are gate, drain, source, and body (bulk). The gate-to-source voltage controls formation of a conducting channel.

An NMOS transistor generally turns ON when its gate-to-source voltage exceeds its positive threshold voltage. In a CMOS inverter, NMOS acts as the pull-down transistor, connecting the output towards ground when the input is HIGH.

A PMOS transistor generally turns ON when its gate voltage is sufficiently lower than its source voltage. In a CMOS inverter, PMOS acts as the pull-up transistor, connecting the output towards the positive supply when the input is LOW.

CMOS means Complementary Metal-Oxide-Semiconductor. The PMOS and NMOS devices respond complementarily to the same input. In ideal stable conditions, one device is OFF, so there is no continuous direct conducting path from supply to ground through the main switching network. Practical circuits still consume power due to leakage and switching activity.

7. CIRCUIT ARCHITECTURE

The CMOS inverter consists of two MOSFETs connected between the positive supply and ground. The PMOS source and body are connected to VDD. The NMOS source and body are connected to ground. Both gates connect to VIN, and both drains connect to VOUT.

When VIN is LOW, PMOS conducts and pulls VOUT towards VDD, while NMOS is OFF. When VIN is HIGH, PMOS is OFF and NMOS conducts, pulling VOUT towards ground. The output is therefore the logical complement of the input.

Conceptual connection: VDD (+5 V) → PMOS → VOUT → NMOS → GND. VIN connects to both transistor gates. Insert your actual LTspice schematic screenshot if available.

8. SOFTWARE AND COMPONENTS

LTspice is an electronic circuit simulation tool supporting schematic capture, SPICE netlists, device models, transient analysis, AC analysis, and DC operating-point analysis. It is used to create the circuit, configure sources and MOSFET models, run transient analysis, and display voltage waveforms.

PMOS forms the pull-up network; NMOS forms the pull-down network. A 5 V DC source provides the supply, and a pulsed voltage source switches the input between 0 V and 5 V. Ground establishes the

reference potential. The simulator calculates node voltages according to the connections, device models, sources, and analysis settings.

9. DESIGN METHODOLOGY

The design begins by identifying the required NOT function, Y = NOT(A). One PMOS and one NMOS transistor are selected. PMOS connects to the positive supply and NMOS to ground. Their gates share the input node and their drains share the output node.

A 5 V DC source is configured for the supply. A pulsed source alternates between 0 V and 5 V. Transient analysis is configured to display several input transitions. After simulation, VIN and VOUT are plotted and compared with the truth table. The expected result is that VOUT assumes the opposite logic state from VIN after each transition and settling interval.

10. SIMULATION PARAMETERS

Supply voltage: 5 V. Input LOW: 0 V. Input HIGH: 5 V. Input rise time: 1 ns. Input fall time: 1 ns. HIGH duration: 5 ms. Pulse period: 10 ms. Transient stop time: 20 ms. Maximum time step: 10 microseconds. NMOS and PMOS dimensions in the example are L = 1 µm and W = 10 µm.

Input source: PULSE(0 5 0 1n 1n 5m 10m)
Transient directive: .tran 0 20m 0 10u

Simplified model definitions:
.model NM NMOS (LEVEL=1 VTO=0.7 KP=120u)
.model PM PMOS (LEVEL=1 VTO=-0.7 KP=50u)

These are educational Level-1 models, not calibrated models for a specific fabrication process. Actual performance depends on model choice, loading, and simulation setup.

11. WORKING PRINCIPLE

Case 1 — Input LOW (0 V): The PMOS source is at 5 V and its gate is at 0 V, so VGS = –5 V. The PMOS turns ON under the simplified model. The NMOS gate and source are both at 0 V, so VGS = 0 V and NMOS is OFF. The output is pulled towards 5 V and represents logic HIGH.

Case 2 — Input HIGH (5 V): The PMOS gate and source are both at 5 V, so VGS = 0 V and PMOS is OFF. The NMOS gate is at 5 V and its source is at ground, so VGS = 5 V. NMOS turns ON and pulls the output towards 0 V, representing logic LOW.

During transitions, one device turns OFF while the other turns ON. Real circuits have finite transition times because of transistor characteristics and capacitances. Both transistors may briefly conduct during switching, causing short-duration current and dynamic power consumption.

12. TRUTH TABLE

The Boolean function is Y = NOT(A), meaning that output is the complement of input.

Input A Output Y
0 1
1 0

For this 5 V simulation, an input near 0 V should produce output near 5 V, and an input near 5 V should produce output near 0 V. Exact values depend on the model and circuit conditions. This truth table is the principal functional criterion for verification.

13. TRANSIENT WAVEFORM ANALYSIS

Transient analysis calculates how voltages and currents change over time. The pulse input alternates between LOW and HIGH. VIN should switch between approximately 0 V and 5 V, while VOUT should switch between approximately 5 V and 0 V. The waveforms should be logical complements.

The output transition may occur slightly after the input transition because of propagation delay and capacitance. Display both VIN and VOUT in the LTspice waveform viewer. Insert the actual waveform screenshot from your simulation if available. Do not label an illustrative waveform as measured simulation data.

Propagation delay is commonly measured between the 50% voltage points of input and output transitions. For a 5 V supply, the midpoint is 2.5 V. The input LOW-to-HIGH/output HIGH-to-LOW delay is tPHL; the input HIGH-to-LOW/output LOW-to-HIGH delay is tPLH. Average delay is tP = (tPHL + tPLH)/2. Report numerical values only after measuring the actual waveform.

14. RESULTS AND DISCUSSION

The expected simulation demonstrates NOT logic using complementary MOSFET switching. For a LOW input, PMOS provides the pull-up path and the output settles towards the positive supply. For a HIGH input, NMOS provides the pull-down path and the output settles towards ground.

Expected observations are: (1) input alternates between 0 V and 5 V; (2) output alternates between HIGH and LOW; (3) output logic complements the input; (4) PMOS acts as the pull-up device; and (5) NMOS acts as the pull-down device. Compare the actual waveform with the truth table before submission. Do not claim measured values that have not been obtained from the simulation.

15. ADVANTAGES

Low static power: In ideal stable states, one transistor is OFF, limiting direct current through the main switching path. Real circuits still have leakage and dynamic power.

High input impedance: The insulated MOSFET gate draws very little steady-state current in the ideal model.

Logic restoration: The inverter can produce clear digital output levels when operated within its intended voltage range.

Compact implementation: A basic CMOS inverter requires only two transistors.

Wide applicability: CMOS structures are used throughout processors, memories, controllers, and digital communication circuits.

Simulation accessibility: LTspice lets students examine circuit behaviour before building hardware.

16. APPLICATIONS

CMOS inverters reverse logic signals and can help restore digital signal levels. They are building blocks inside microprocessors, microcontrollers, memory circuits, and digital signal-processing systems. Cross-coupled inverter structures are used in SRAM cells. Inverters can be connected with feedback networks to create oscillator circuits, including ring oscillators. In communication and embedded systems, inverter stages appear in timing, control, and signal-conditioning blocks.

17. LIMITATIONS

Level-1 MOSFET models are simplified and do not capture all short-channel effects, parasitic capacitances, process variations, or advanced semiconductor behaviour. The project is simulation-only; no physical integrated circuit has been fabricated or measured. The basic analysis does not automatically provide accurate delay, power, or noise-margin values. The design uses a fixed 5 V supply, while practical CMOS technologies may use different supplies and process-specific models. Load capacitance affects transition times. Physical layout, design-rule checking, layout-versus-schematic verification, and parasitic extraction are outside the scope.

18. TROUBLESHOOTING AND VERIFICATION

If the output remains constant, check whether the input waveform changes, both transistor gates share VIN, and both drains connect to VOUT. Confirm PMOS source is connected to VDD and NMOS source to ground. Ensure a ground reference exists.

A floating-node warning can indicate a missing connection or an unreferenced node. Check wiring and node labels. If output levels are incorrect, verify MOSFET terminal order and model names. In the netlist format used here, the MOSFET terminal order is drain, gate, source, bulk. If no transitions appear, check pulse parameters and transient stop time. Plot VIN and VOUT together to verify inversion.

19. FUTURE SCOPE

The project can be extended through DC transfer-characteristic analysis, plotting output voltage against input voltage to study switching threshold and noise margins. Propagation delay can be measured using the 50% points of transitions. Supply current can be examined to estimate power, distinguishing static and dynamic components. Varying transistor dimensions can show the effect of sizing. Further projects can implement CMOS NAND, NOR, XOR, and XNOR gates. An advanced flow could include layout creation, design-rule checking, layout-versus-schematic verification, and parasitic extraction.

20. CONCLUSION

This project presents the design and simulation of a basic CMOS inverter using LTspice. One PMOS transistor acts as the pull-up device and one NMOS transistor acts as the pull-down device. Their gates receive the same input and their drains share the output node. Complementary switching causes the output to represent the logical complement of the input.

With a 5 V supply and pulsed input between 0 V and 5 V, transient analysis demonstrates the expected inverter truth table. The exercise develops practical understanding of transistor-level connections, SPICE simulation, and waveform interpretation. It provides a foundation for more complex CMOS logic design. Results must be interpreted within the limits of simplified educational models and should not be treated as physical chip measurements.

21. REFERENCES

1. Analog Devices, LTspice documentation and built-in help resources.
2. Neil H. E. Weste and David Money Harris, CMOS VLSI Design: A Circuits and Systems Perspective, Pearson.
3. Jan M. Rabaey, Anantha Chandrakasan, and Borivoje Nikoli■, Digital Integrated Circuits: A Design Perspective, Pearson.
4. Adel S. Sedra and Kenneth C. Smith, Microelectronic Circuits, Oxford University Press.
5. Course notes and instructional materials provided as part of the Glow Logics VLSI training programme.

APPENDIX A: LTSPICE NETLIST

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* CMOS inverter - basic LTspice simulation
VDD vdd 0 DC 5
VIN in 0 PULSE(0 5 0 1n 1n 5m 10m)
M1 out in vdd vdd PM L=1u W=10u
M2 out in 0 0 NM L=1u W=10u
.model NM NMOS (LEVEL=1 VTO=0.7 KP=120u)
.model PM PMOS (LEVEL=1 VTO=-0.7 KP=50u)
.tran 0 20m 0 10u
.end
```

APPENDIX B: FIGURES AND SUBMISSION

Figure 1: Insert the actual LTspice CMOS inverter schematic screenshot if available.
Figure 2: Insert the actual LTspice transient waveform showing VIN and VOUT if available.

Do not label illustrative graphics as actual simulation results. Before submission, verify the schematic, include actual screenshots if possible, check that the truth table matches the waveform, and include the schematic, netlist, report, screenshots, and README.md in the public GitHub repository if required by the course.